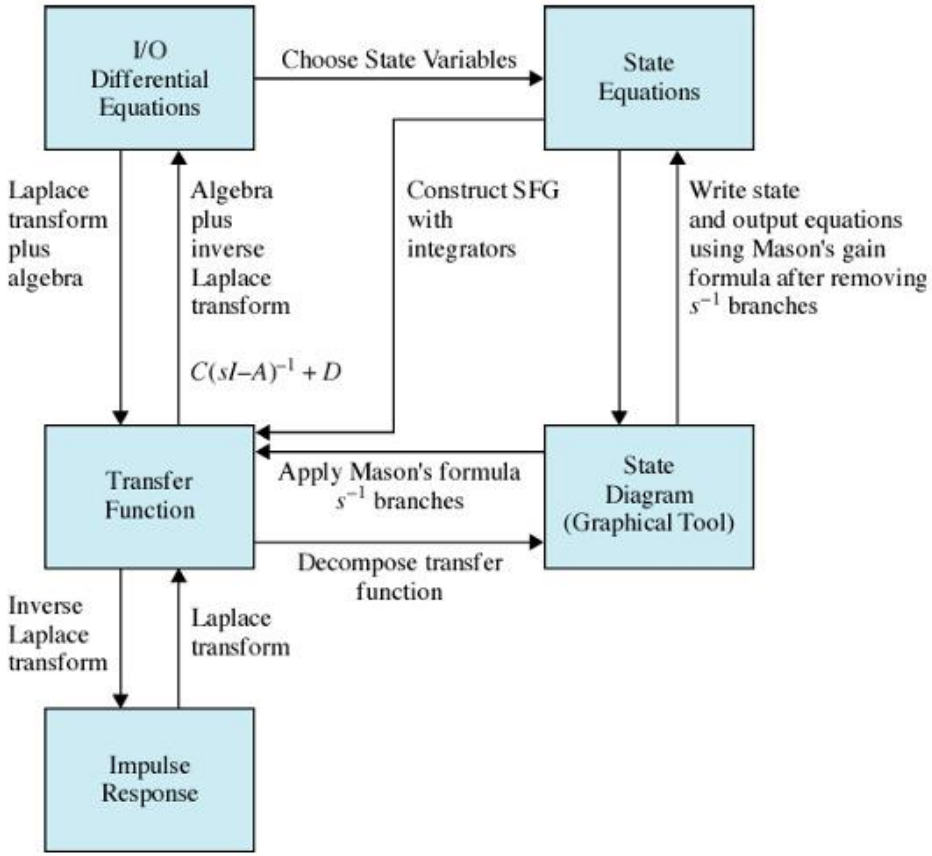
제어공학2
Automatic Control

State-Space Analysis and Controller Design

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전달함수의 분해

✓선형시스템을 기술하는 여러 방법 사이의 상호관계



전달함수의 분해

✓CCF로의 직접분해

- 전달함수를 s의 음의 멱급수로 표현, 분모, 분자에 s^{-n}을 곱함
- 전달함수 분모, 분자에 가상변수 X(s)를 곱함
- 분모, 분자를 각각 서로 같게 놓음

$$\frac{Y(s)}{U(s)}=\frac{b_{n-1}s^{n-1}+b_{b-2}s^{n-2}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$$

$$\frac{Y(s)}{U(s)}=\frac{b_{n-1}s^{-1}+b_{n-2}s^{-2}+\cdots+b_1s^{-n+1}+b_0s^{-n}}{1+a_{n-1}s^{-1}+\cdots+a_1s^{-n+1}+a_0s^{-n}}\frac{X(s)}{X(s)}$$

$$Y(s)=\left(b_{n-1}s^{-1}+b_{n-2}s^{-2}+\cdots+b_1s^{-n+1}+b_0s^{-n}\right)X(s)$$

$$U(s)=\left(1+a_{n-1}s^{-1}+\cdots+a_1s^{-n+1}+a_0s^{-n}\right)X(s)$$

$$X(s)=U(s)-\left(a_{n-1}s^{-1}+a_{n-2}s^{-2}+\cdots+a_1s^{-n+1}+a_0s^{-n}\right)X(s)$$

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

$$y(t)=\mathbf{C}\mathbf{x}(t)+Du(t)$$

$$\mathbf{A}=\begin{bmatrix}0&1&0&\cdots&0\\0&0&1&\cdots&0\\\vdots&\vdots&\vdots&\ddots&\vdots\\0&0&0&0&1\\-a_0&-a_1&-a_2&\cdots&-a_{n-1}\end{bmatrix}\quad\mathbf{B}=\begin{bmatrix}0\\0\\\vdots\\0\\1\end{bmatrix}$$

$$\mathbf{C}=\begin{bmatrix}b_0&b_1&\cdots&b_{n-2}&b_{n-1}\end{bmatrix}\quad\mathbf{D}=0$$

전달함수의 분해

✓CCF로의 직접분해

$$\frac{Y(s)}{U(s)}=\frac{b_{n-1}s^{n-1}+b_{b-2}s^{n-2}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$$

$$\frac{Y(s)}{U(s)}=\frac{2s^2+s+5}{s^3+6s^2+11s+4}$$

$$Y(s)=(2s^{-1}+s^{-2}+5s^{-3})X(s)$$

$$X(s)=U(s)-(6s^{-1}+11s^{-2}+4s^{-3})X(s)$$

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

$$y(t)=\mathbf{C}\mathbf{x}(t)+Du(t)$$

$$\begin{bmatrix}\frac{dx_1(t)}{dt}\\ \frac{dx_2(t)}{dt}\\ \frac{dx_3(t)}{dt}\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -4 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\\ x_3(t)\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u(t)$$

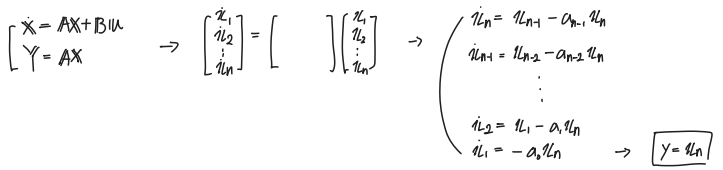
$$y(t)=[5\quad 1\quad 2]x(t)$$

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전달함수의 분해

✓OCF로의 직접분해

- 분모, 분자에 s^{-n}을 곱하면



$$\frac{Y(s)}{U(s)}=\frac{b_{n-1}s^{n-1}+b_{b-2}s^{n-2}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$$

$$\left(1+a_{n-1}s^{-1}+\cdots+a_1s^{-n+1}+a_0s^{-n}\right)Y(s)=\left(b_{n-1}s^{-1}+b_{n-2}s^{-2}+\cdots+b_1s^{-n+1}+b_0s^{-n}\right)U(s)$$

$$Y(s)=-\left(a_{n-1}s^{-1}+\cdots+a_1s^{-n+1}+a_0s^{-n}\right)Y(s)+\left(b_{n-1}s^{-1}+b_{n-2}s^{-2}+\cdots+b_1s^{-n+1}+b_0s^{-n}\right)U(s)$$

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

$$y(t)=\mathbf{C}\mathbf{x}(t)+Du(t)$$

$$\mathbf{A}=\begin{bmatrix}0&0&\cdots&0&-a_0\\1&0&\cdots&0&-a_1\\0&1&\cdots&0&-a_2\\\vdots&\vdots&\ddots&\vdots&\vdots\\0&0&\cdots&1&-a_{n-1}\end{bmatrix}\quad\mathbf{B}=\begin{bmatrix}b_0\\b_1\\b_2\\\vdots\\b_{n-1}\end{bmatrix}\quad\mathbf{C}=\begin{bmatrix}0&0&\cdots&0&1\end{bmatrix}\quad\mathbf{D}=0$$

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전달함수의 분해

✓OCF로의 직접분해

$$\frac{Y(s)}{U(s)}=\frac{b_{n-1}s^{n-1}+b_{b-2}s^{n-2}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$$

$$\frac{Y(s)}{U(s)}=\frac{2s^2+s+5}{s^3+6s^2+11s+4}$$

$$Y(s)=(2s^{-1}+s^{-2}+5s^{-3})U(s)-(6s^{-1}+11s^{-2}+4s^{-3})Y(s)$$

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

$$y(t)=\mathbf{C}\mathbf{x}(t)+Du(t)$$

$$\begin{bmatrix}\frac{dx_1(t)}{dt}\\ \frac{dx_2(t)}{dt}\\ \frac{dx_3(t)}{dt}\end{bmatrix}=\begin{bmatrix}0&0&-4\\1&0&-11\\0&1&-6\end{bmatrix}\begin{bmatrix}x_1(t)\\x_2(t)\\x_3(t)\end{bmatrix}+\begin{bmatrix}5\\1\\2\end{bmatrix}u(t)$$

$$y(t)=\begin{bmatrix}0&0&1\end{bmatrix}\mathbf{x}(t)$$

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제어시스템의 가제어성 Controllability of control system

- ✓상태 피드백 State feedback
- ✓극배치 설계 Pole-placement design

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{u}(t) = -\mathbf{K}\mathbf{x}(t) + \mathbf{r}(t)$$

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}\mathbf{r}(t)$$

$u = r - kx$

$\dot{x} = Ax + Br - BkX$

$= (A - Bk)x + Br$

$= A$

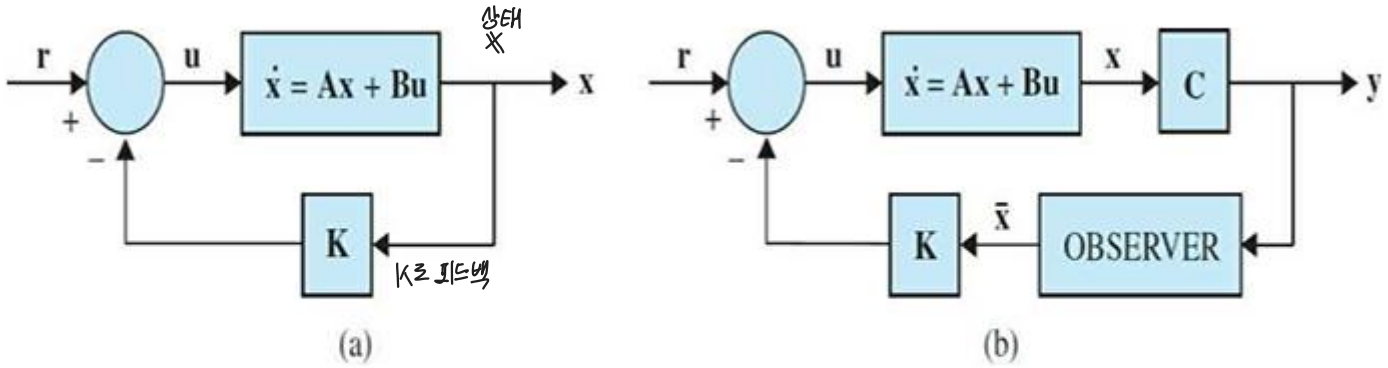
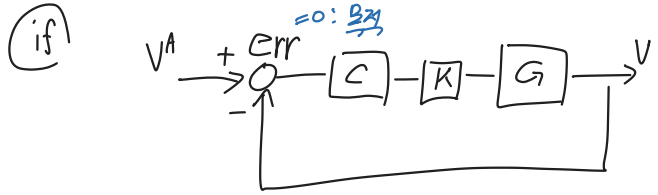


Figure 8-16 (*a*) Control system with state feedback. (*b*) Control system with observer and state feedback.

제어시스템의 가제어성 Controllability of control system

✓가제어성의 일반개념

- 모든 상태변수가 제어입력 u(t)에 의해 유한시간 내에 일정 목표값에 도달할 수 있다면 완전 제어가능 (completely controllable)



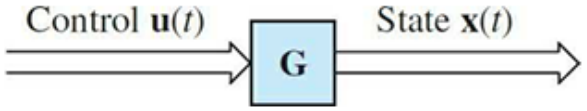
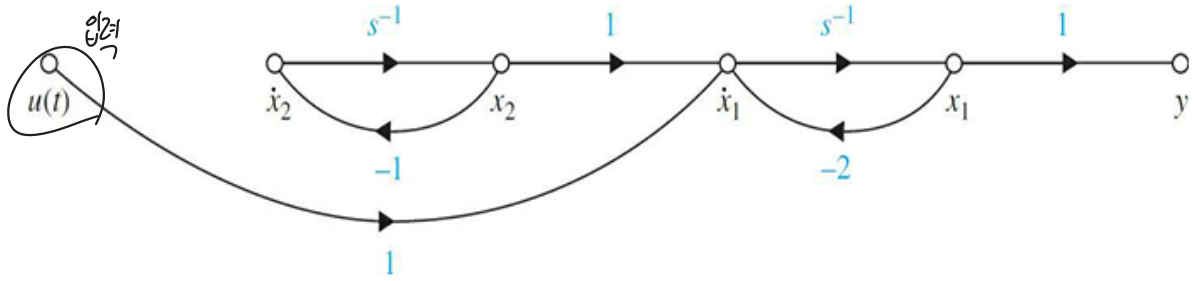


Figure 8-17 Linear time-invariant system.



$\dot{x}_1 = -2x_1 + u$

$\dot{x}_2 = -x_2$ (if $\dot{x}_2 = +x_2$: 제어불가)

$c \cdot e^{-t}$

Figure 8-18 State diagram of the system that is not state controllable. : 입력의 변동이 있어도 x_2에 영향이 가지 않음.

제어시스템의 가제어성 Controllability of control system

✓상태 가제어성의 정의 Definition of state controllability

- 가제어성 행렬의 rank가 full rank이면 완전한 상태 가제어성이 보장

가 제어성 행렬의 역행렬 존재 유무로 제어 유무를 판단할 수 있다.

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\mathbf{A} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix}$$

여부의 역행렬 유무로 제어성판별

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

if 일때,

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \quad : rank = 1$$

?필기실패_



M은 모두건 역행렬 존재

제어시스템의 가제어성 Controllability of control system

✓상태 가제어성의 정의 Definition of state controllability

- 가제어성 행렬의 rank가 full rank이면 완전한 상태 가제어성이 보장

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & -4 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \end{bmatrix} = \begin{bmatrix} 0 & -1 & -4 \\ 0 & 0 & 0 \\ 1 & 3 & 8 \end{bmatrix}$$

선형시스템의 가관측성 Observability of linear system

✓가관측성의 정의 Definition of observability

- 가관측성 행렬의 rank가 full rank이면 완전한 상태 가제어성이 보장

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

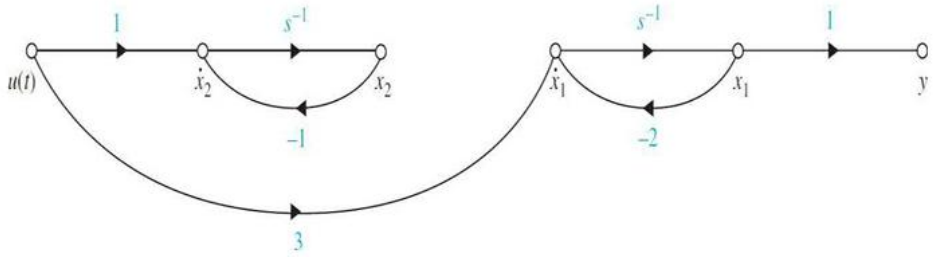
$$\mathbf{A} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

여의 역행렬 유무로 관측성판별 $\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix}$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$$

1차독

이것만 독립 ⇒ rank=1



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가제어성, 가관측성과 전달함수 사이의 관계

✓가제어성, 가관측성

- 입출력 전달함수에 극-영점 상쇄가 일어나지 않는다면, 이 시스템은 항상 가제어/가관측 시스템인 동적방정식으로 표현할 수 있음

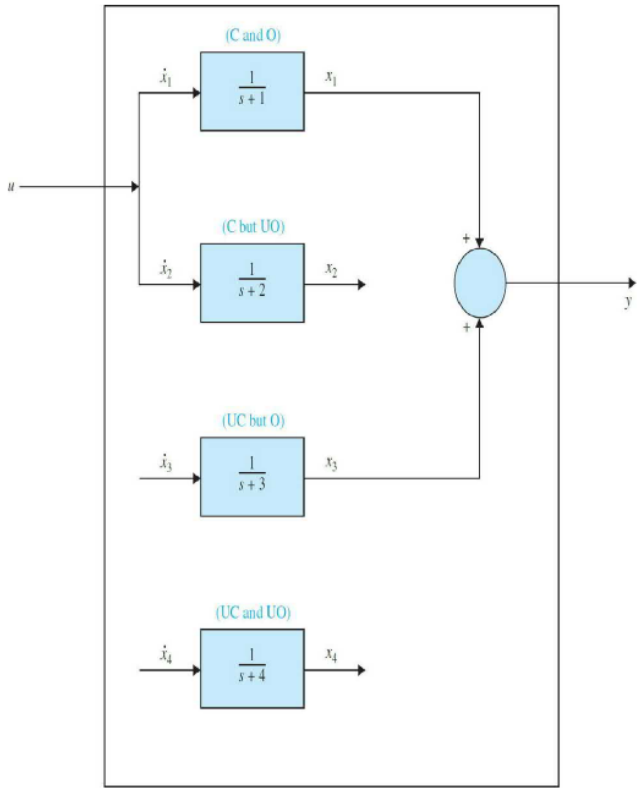
$$A=\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix} \quad B=\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad C=\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix} \quad D=0$$

x_1: Controllable and observable (C and O)

x_2: Controllable but unobservable (C but UO)

x_3: Uncontrollable but observable (UC but O)

x_4: Uncontrollable and unobservable (UC and UO)



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가제어성, 가관측성과 전달함수 사이의 관계

✓가제어성, 가관측성

$$\frac{Y(s)}{U(s)} = \frac{s+2}{(s+1)(s+2)} = \frac{s+2}{s^2+3s+2}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix}$$

: 제어가능
but, 관측불가

$$\mathbf{A} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}$$

: 역행렬 없음
→ 관측 가능
but, 제어불가

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가제어성과 가관측성에 관한 불변정리

✓유사변환에 관한 불변정리

- 가제어성과 가관측성은 유사변환에 영향을 받지 않음

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \bar{\mathbf{A}}\bar{\mathbf{x}}(t) + \bar{\mathbf{B}}\mathbf{u}(t)$$

$$\bar{\mathbf{y}}(t) = \bar{\mathbf{C}}\mathbf{x}(t) + \bar{\mathbf{D}}\mathbf{u}(t)$$

$$\bar{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \quad \bar{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

$$\bar{S} = \begin{bmatrix} P^{-1}B & P^{-1}AP P^{-1}B & \cdots \end{bmatrix}$$

$$= \begin{bmatrix} P^{-1}B & P^{-1}AB & \cdots \end{bmatrix}$$

$$= P^{-1}\begin{bmatrix} B & AB & \cdots \end{bmatrix}$$

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가제어성과 가관측성에 관한 불변정리

✓상태 피드백을 갖는 폐루프 시스템의 가제어성에 관한 정리

- 완전제어가능이면 상태 피드백을 갖는 폐루프 시스템도 완전히 제어가능
- 만약 제어불가능하면 제어가능하게 만들 수 있는 K가 존재하지 않음

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{u}(t) = \mathbf{r}(t) - \mathbf{K}\mathbf{x}(t)$$

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}\mathbf{r}(t)$$

제어유막 : A, B를 포함
관측유막 : A, C "

$S = [B \quad AB \quad \cdots]$

$\bar{S} = [B \quad (A - BK) \quad \cdots]$

$= [B \quad AB - BKB \quad \cdots]$

α ⟶ Span (상식배 이므로 □ 가 변하지 않음)

$[B \quad (A - \alpha I)B]$

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가제어성과 가관측성에 관한 불변정리

✓상태 피드백을 갖는 폐루프 시스템의 가관측성에 관한 정리

- 만약 개루프시스템이 가제어이고 가관측이면 상태피드백을 갖는 폐루프 시스템은 가관측성을 잃을 수 있음

$$\mathbf{A}=\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad \mathbf{B}=\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{C}=\begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$u(t)=r(t)-\mathbf{K}\mathbf{x}(t)$$

$$\mathbf{K}=[k_1 \quad k_2]$$

$$\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}k_1 & k_2\end{bmatrix}=\begin{bmatrix}k_1 & k_2\\ k_1 & k_2\end{bmatrix}$$

$$\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}-\begin{bmatrix}k_1 & k_2\\ k_1 & k_2\end{bmatrix}$$

$$\frac{d\mathbf{x}(t)}{dt}=(\mathbf{A}-\mathbf{BK})\mathbf{x}(t)+\mathbf{B}\mathbf{r}(t)$$

$$\mathbf{A}-\mathbf{BK}=\begin{bmatrix} -k_1 & 1-k_2 \\ -2-k_1 & -3-k_2 \end{bmatrix}$$

$$\mathbf{V}=\begin{bmatrix} \mathbf{C} \\ \mathbf{C}(\mathbf{A}-\mathbf{BK}) \end{bmatrix}=\begin{bmatrix} 1 & 2 \\ -3k_1-4 & -3k_2-5 \end{bmatrix}$$

$$\begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}-k_1 & 1-k_2\\ -2-k_1 & -3-k_2\end{bmatrix}=\begin{bmatrix}1 & 2\\ -3k_1-4 & -3k_2-5\end{bmatrix}=V$$

$$\therefore -3k_2-5+2(3k_1+4)=6k_1-3k_2+3$$

$$|\mathbf{V}|=6k_1-3k_2+3$$

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전자석-쇠구슬 현수 시스템 Magnetic-ball suspension system

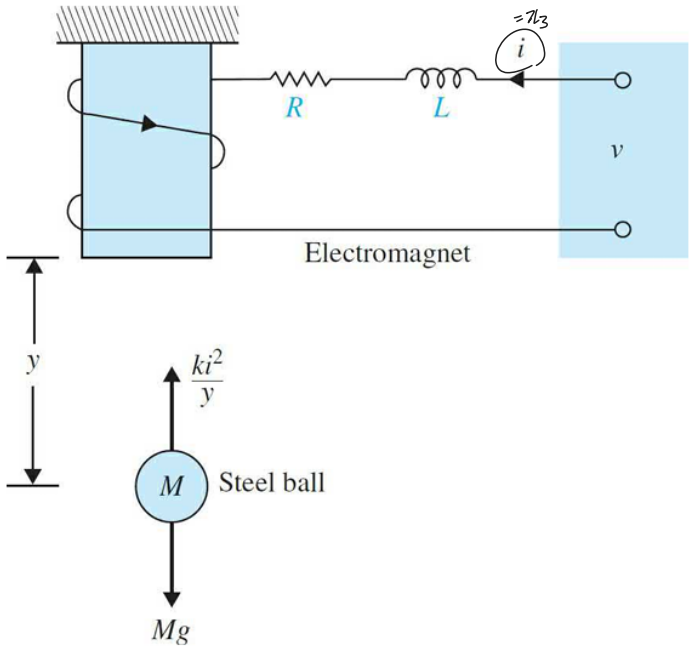


Figure 8-21 Ball-suspension system.

$$M\frac{d^2y(t)}{dt^2} = Mg - \frac{ki^2(t)}{x(t)}$$

$$v(t) = Ri(t) + L\frac{di(t)}{dt}$$

$$x_1(t) = x(t)$$

$$x_2(t) = \frac{dx(t)}{dt}$$

$$x_3(t) = i(t)$$

$$\frac{dx_1(t)}{dt} = x_2(t)$$

$$\frac{dx_2(t)}{dt} = g - \frac{k}{M}\frac{x_3^2(t)}{x_1(t)}$$

$$\frac{dx_3(t)}{dt} = -\frac{R}{L}x_3(t) + \frac{v(t)}{L}$$

$v(t)$ = input voltage (V)

$i(t)$ = winding current (A)

R = winding resistance = 1 Ω

M = ball mass = 1.0 kg

$x(t)$ = ball position (m)

k = proportional constant = 1.0

L = winding inductance = 0.01 H

g = gravitational acceleration = 32.2 m/s^2

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전자석-쇠구슬 현수 시스템 Magnetic-ball suspension system

$$M\frac{d^2y(t)}{dt^2} = Mg - \frac{ki^2(t)}{x(t)}$$

$$v(t) = Ri(t) + L\frac{di(t)}{dt}$$

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{A}^*\,\Delta\mathbf{x}(t) + \mathbf{B}^*\,\Delta v(t)$$

$$y_0(t) = x_{01} = 0.5\ \text{m}$$

$$\mathbf{A}^* = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \cdot & \cdot & \cdots & \cdot \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} \frac{\partial f_1}{\partial r_1} & \frac{\partial f_1}{\partial r_2} & \cdots & \frac{\partial f_1}{\partial r_p} \\ \frac{\partial f_2}{\partial r_1} & \frac{\partial f_2}{\partial r_2} & \cdots & \frac{\partial f_2}{\partial r_p} \\ \cdot & \cdot & \cdots & \cdot \\ \frac{\partial f_n}{\partial r_1} & \frac{\partial f_n}{\partial r_2} & \cdots & \frac{\partial f_n}{\partial r_p} \end{bmatrix}$$

$$x_1(t) = x(t) \qquad \frac{dx_1(t)}{dt} = x_2(t)$$

$$x_2(t) = \frac{dx(t)}{dt} \qquad \frac{dx_2(t)}{dt} = g - \frac{k}{M}\frac{x_3^2(t)}{x_1(t)}$$

$$x_3(t) = i(t) \qquad \frac{dx_3(t)}{dt} = -\frac{R}{L}x_3(t) + \frac{v(t)}{L}$$

$$\begin{bmatrix} \delta\dot{x}_1 \\ \delta\dot{x}_2 \\ \delta\dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 2g & 0 & -2\sqrt{\frac{2kg}{M}} \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \delta x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix} \delta v \qquad \delta y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \delta x_3 \end{bmatrix}$$

$v(t)$ = input voltage (V)

$i(t)$ = winding current (A)

R = winding resistance = 1 Ω

M = ball mass = 1.0 kg

$x(t)$ = ball position (m)

k = proportional constant = 1.0

L = winding inductance = 0.01 H

g = gravitational acceleration = 32.2 m/s^2

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}$$

전자석-쇠구슬 현수 시스템 Magnetic-ball suspension system

✓특성방정식, 고유값, 전달함수, 가제어성

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}$$

$$\frac{Y(s)}{V(s)} = \mathbf{C}^*(s\mathbf{I}-\mathbf{A}^*)^{-1}\mathbf{B}^* = [1 \;\; 0 \;\; 0](s\mathbf{I}-\mathbf{A}^*)^{-1}\mathbf{B}^* = \frac{-1600}{(s+100)(s+8.025)(s-8.025)}$$

$$|s\mathbf{I}-\mathbf{A}^*| = \begin{bmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 0 & 0 & s+100 \end{bmatrix} = s^3+100s^2-64.4s-6440 = 0$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{B}^* & \mathbf{A}^*\mathbf{B}^* & \mathbf{A}^{*2}\mathbf{B}^* \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1,600 \\ 0 & -1,600 & 160,000 \\ 100 & -10,000 & 1,000,000 \end{bmatrix}$$

But, 역행렬 존재 → 제어가능

$s = -100, -8.025, 8.025$ pole이 우반면 위치 ⇒ unstable system

전자석-쇠구슬 현수 시스템 Magnetic-ball suspension system

✓가관측성

출력을 볼로 잡냐에 따라 관측성이 변할 수 있음.

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}$$

거리출력 → 모든상태 관측 가능

$$\mathbf{C}^* = [1\ 0\ 0]$$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^{*2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 64.4 & 0 & -16 \end{bmatrix}$$

속도출력 → 나머지상태 관측 가능

$$\mathbf{C}^* = [0\ 1\ 0]$$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^{*2} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 64.4 & 1600 \end{bmatrix}$$

rank=3.

전류출력 → 불가능

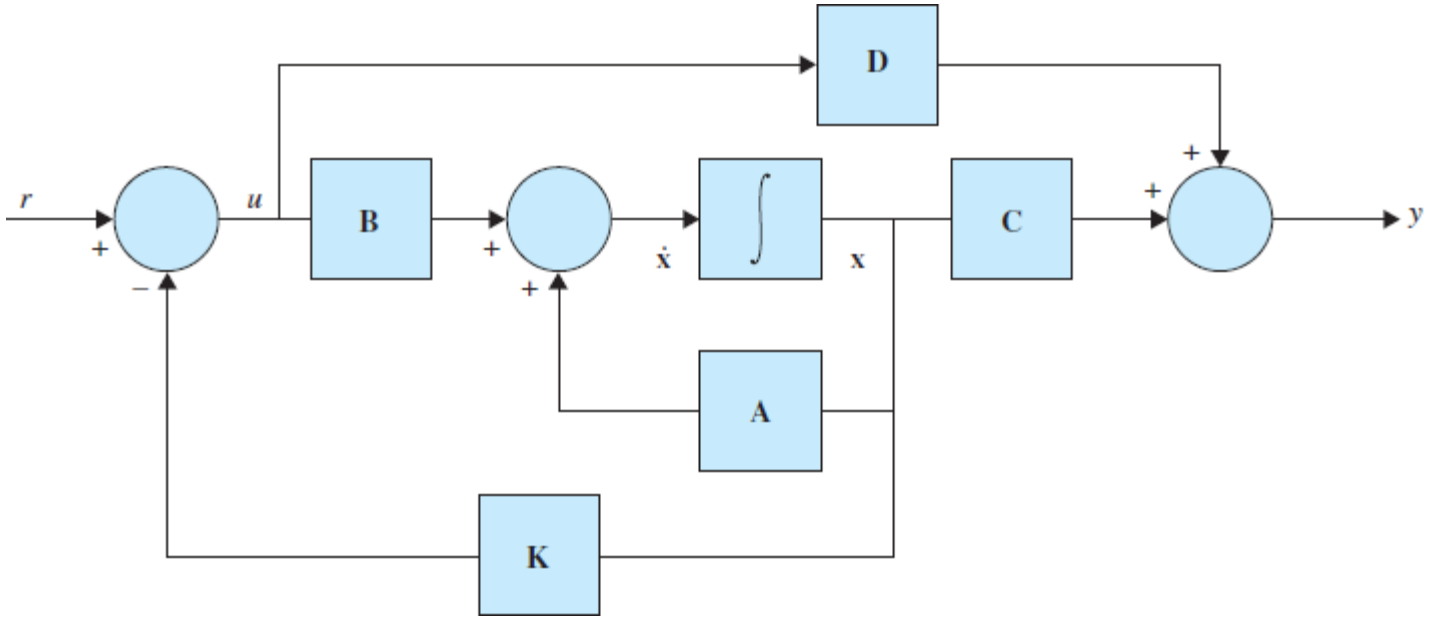
$$\mathbf{C}^* = [0,0,1]$$

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^{*2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -100 \\ 0 & 0 & -10{,}000 \end{bmatrix}$$

상태피드백 제어 State-feedback control

✓State-feedback

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\zeta\omega_n + K_2)s + K_1}$$



- PD control

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2(K_P + K_D s)}{s^2 + (2\zeta\omega + K_D\omega_n^2)s + \omega_n^2 K_P}$$

상태피드백을 이용한 극배치 설계 Pole-placement design

✓3차 이상의 제어 시스템

- PD, PI, 1단 진상 또는 지상제어기는 독립적으로 모든 극들을 제어할 수 없음
- 상태피드백을 이용하면 n개의 고유값들을 임의로 배치할 수 있음

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$u(t) = -\mathbf{K}\mathbf{x}(t) + r(t)$$

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}r(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \ddots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{A} - \mathbf{B}\mathbf{K} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 - k_1 & -a_1 - k_2 & -a_2 - k_3 & \cdots & -a_{n-1} - k_n \end{bmatrix}$$

$$\mathbf{K} = [k_1 \quad k_2 \quad \cdots \quad k_n]$$

$$|s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K}| = 0$$

$$|s\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K})| = s^n + (a_{n-1} + k_n)s^{n-1} + (a_{n-2} + k_{n-1})s^{n-2} + \cdots + (a_0 + k_1) = 0$$

상태피드백을 이용한 극배치 설계 Pole-placement design

$$\frac{d\Delta\mathbf{x}(t)}{dt} = \mathbf{A}^*\,\Delta\mathbf{x}(t) + \mathbf{B}^*\,\Delta v(t)$$

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix} \quad [k_1\ k_2\ k_3]$$

$s = -100, -8.025, 8.025$ ← 이 상태였는데 k를 잘 써서

$$\Delta v(t) = -\mathbf{K}\Delta\mathbf{x}(t) = -[k_1 \quad k_2 \quad k_3]\Delta\mathbf{x}(t)$$

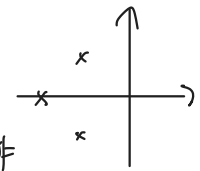
$$s\mathbf{I} - \mathbf{A}^* + \mathbf{B}^*\mathbf{K} = \begin{vmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 100k_1 & 100k_2 & s+100+100k_3 \end{vmatrix}$$

$$= s^3 + 100(k_3+1)s^2 - (64.4+1600k_2)s - 1600k_1 - 6440(k_3+1) = 0$$

1. The system must be stable.
2. For any initial disturbance on the position of the ball from its equilibrium position, the ball must return to the equilibrium position with zero steady-state error.
3. The time response should settle to within 5 percent of the initial disturbance in not more than 0.5 s.
4. The control is to be realized by state feedback

→ $s = -20\ s = -6+j4.9\ s = -6-j4.9$: pole이 어디 위치해야 위 조건을 모두 만족할 수 있음.

pole을 내가 원하는 위치로 바꿈. ★★ 핵심.



$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} = \begin{bmatrix} -2.038 & -0.22775 & -0.68 \end{bmatrix}$$

상태피드백을 이용한 극배치 설계 Pole-placement design

$$\frac{d\Delta\mathbf{x}(t)}{dt} = \mathbf{A}^*\,\Delta\mathbf{x}(t) + \mathbf{B}^*\,\Delta v(t)$$

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}$$

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$s = -100, -8.025, 8.025$$

$$\Delta v(t) = -\mathbf{K}\Delta\mathbf{x}(t) = -[k_1 \quad k_2 \quad k_3]\Delta\mathbf{x}(t)$$

$$s\mathbf{I} - \mathbf{A}^* + \mathbf{B}^*\mathbf{K} = \begin{vmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 100k_1 & 100k_2 & s+100+100k_3 \end{vmatrix}$$

$$= s^3 + 100(k_3+1)s^2 - (64.4+1600k_2)s - 1600k_1 - 6440(k_3+1) = 0$$

$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} = \begin{bmatrix} -2.038 & -0.22775 & -0.68 \end{bmatrix}$$

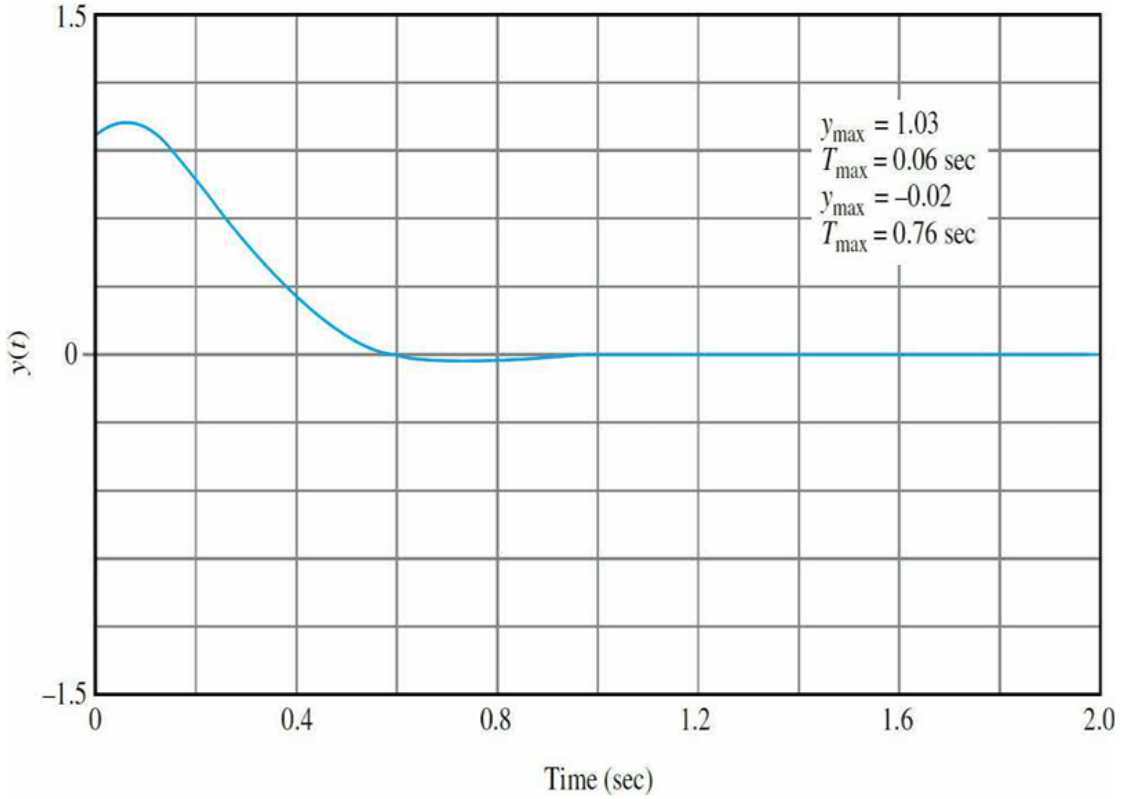


Figure 8-25 Output response of magnetic-ball-suspension system with state feedback, subject to initial condition $y(0) = x_1(0) = 1$.

상태피드백을 이용한 극배치 설계 Pole-placement design

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\theta_e(t)$$

$$\theta_e(t) = -\mathbf{K}\mathbf{x}(t) = -[k_1 \quad k_2]\mathbf{x}(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -25 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\theta_o(t) = \mathbf{C}\mathbf{x}(t)$$

$$\mathbf{C} = [1 \quad 0]$$

1. The steady-state error due to a step function input should equal 0.
2. With the state-feedback control, the unit-step response should have minimum overshoot, rise time, and settling time.

→ $s = -50$

$$\frac{\Theta_o(s)}{\Theta_r(s)} = \frac{2500}{s^2 + (25 + k_2)s + k_1}$$

$$\mathbf{K} = [2500 \quad 75]$$

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적분제어를 포함한 상태피드백

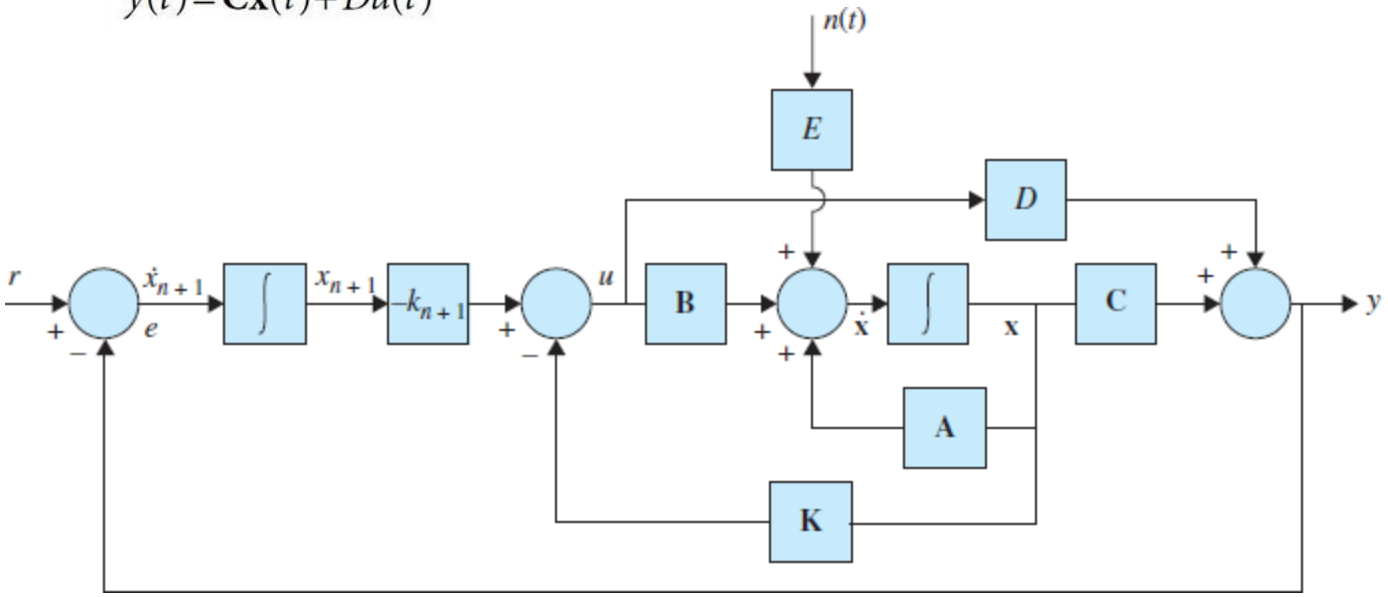
$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{E}n(t)$$

$$\frac{dx_{n+1}(t)}{dt} = r(t) - y(t)$$

$$y(t) = \mathbf{C}\mathbf{x}(t) + Du(t)$$

$$u(t) = -\mathbf{K}\mathbf{x}(t) - k_{n+1}x_{n+1}(t)$$

$$\mathbf{K} = [k_1 \quad k_2 \quad k_3 \quad \cdots \quad k_n]$$



$$\frac{d\bar{\mathbf{x}}(t)}{dt} = (\bar{\mathbf{A}} - \bar{\mathbf{B}}\bar{\mathbf{K}})\bar{\mathbf{x}}(t) + \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} r(t)$$

$$\bar{\mathbf{x}}(t) = \begin{bmatrix} \dfrac{d\mathbf{x}(t)}{dt} \\ \dfrac{dx_{n+1}(t)}{dt} \end{bmatrix} (n+1)\times 1$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & 0 \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} (n+1)\times(n+1) \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ D \end{bmatrix} (n+1)\times 1$$

$$\bar{\mathbf{K}} = \begin{bmatrix} K & K_{n+1} \end{bmatrix} = \begin{bmatrix} k_1 & k_2 & \cdots & k_n & k_{n+1} \end{bmatrix} 1\times(n+1)$$

$$y(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t)$$

$$\bar{\mathbf{C}} = [\mathbf{C} - D\mathbf{K} \quad D\mathbf{K}][1\times(n+1)]$$

적분제어를 포함한 상태피드백

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = (\bar{\mathbf{A}} - \bar{\mathbf{B}}\bar{\mathbf{K}})\bar{\mathbf{x}}(t) + \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} r(t)$$

$$\bar{\mathbf{x}}(t) = \begin{bmatrix} \dfrac{d\mathbf{x}(t)}{dt} \\ \dfrac{dx_{n+1}(t)}{dt} \end{bmatrix} (n+1)\times 1$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & 0 \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} (n+1)\times(n+1) \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ D \end{bmatrix} (n+1)\times 1$$

$$\bar{\mathbf{K}} = \begin{bmatrix} K & K_{n+1} \end{bmatrix} = \begin{bmatrix} k_1 & k_2 & \cdots & k_n & k_{n+1} \end{bmatrix} 1\times(n+1)$$

$$y(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t)$$

$$\bar{\mathbf{C}} = [\mathbf{C} - D\mathbf{K} \quad D\mathbf{K}][1\times(n+1)]$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -25 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 2500 & 0 \end{bmatrix} \quad D = 0$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -25 & 0 \\ -2500 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\left| s\mathbf{I} - \bar{\mathbf{A}} + \bar{\mathbf{B}}\bar{\mathbf{K}} \right| = \begin{vmatrix} s & -1 & 0 \\ k_1 & s+25+k_2 & k_3 \\ -2500 & 0 & s \end{vmatrix}$$

$$= s^3 + (25+k_2)s^2 + k_1 s + 2500k_3 = 0$$

적분제어를 포함한 상태피드백

$$|s\mathbf{I}-\bar{\mathbf{A}}+\bar{\mathbf{B}}\bar{\mathbf{K}}|=\begin{vmatrix} s & -1 & 0 \\ k_1 & s+25+k_2 & k_3 \\ -2500 & 0 & s \end{vmatrix}$$

$$=s^3+(25+k_2)s^2+k_1s+2500k_3=0$$

1. The steady-state output must follow a step function input with zero error.
2. The rise time and settling time must be less than 0.05 s.
3. The maximum overshoot of the unit-step response must be less than 5 percent

→ $s=-200$ $s=-50+j50$ and $-50-j50$

$$s^3+300s^2+25{,}000s+1{,}000{,}000=0$$

$k_1=25{,}000$ $k_2=275$ and $k_3=400$

적분제어를 포함한 상태피드백

$$\frac{d\omega(t)}{dt} = \frac{-B}{J}\omega(t) + \frac{K_i}{J}i_a(t) - \frac{1}{J}T_L$$

$$\frac{di_a(t)}{dt} = \frac{-K_b}{L}\omega(t) - \frac{R}{L}i_a(t) + \frac{1}{L}e_a(t)$$

$$y(t) = \mathbf{C}\mathbf{x}(t) = \begin{bmatrix} 1 & 0 \end{bmatrix}\mathbf{x}(t)$$

$i_a(t)$ = armature current, A

$e_a(t)$ = armature applied voltage, V

$\omega(t)$ = motor velocity, rad/s

B = viscous-friction coefficient of motor and load = 0

J = moment of inertia of motor and load = 0.02 N·m/rad/s^2

K_i = motor torque constant = 1 N·m/A

K_b = motor back-emf constant = 1 V/rad/s

T_L = constant load torque (magnitude not known), N·m

L = armature inductance = 0.005 H

R = armature resistance = 1 Ω

The eigenvalues of the closed-loop system with state feedback and integral control are at $s = -300, -10 + j\,10$, and $-10 - j10$

$$\mathbf{A} = \begin{bmatrix} -\frac{B}{J} & \frac{K_i}{J} \\ -\frac{K_b}{L} & -\frac{R}{L} \end{bmatrix} = \begin{bmatrix} 0 & 50 \\ -200 & -200 \end{bmatrix} \qquad n(t) = T_L u_s(t)$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \end{bmatrix} \qquad \bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & 0 \\ -\mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 50 & 0 \\ -200 & -200 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \\ 0 \end{bmatrix}$$

$$\mathbf{E} = \begin{bmatrix} -\frac{1}{J} \\ 0 \end{bmatrix} = \begin{bmatrix} -50 \\ 0 \end{bmatrix} \qquad \bar{\mathbf{C}} = \begin{bmatrix} \mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{E}} = \begin{bmatrix} \mathbf{E} \\ 0 \end{bmatrix} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix}$$

적분제어를 포함한 상태피드백

$$s = -300, -10 + j\,10, \text{ and } -10 - j10$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & 0 \\ -\mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 50 & 0 \\ -200 & -200 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \\ 0 \end{bmatrix}$$

$$\bar{\mathbf{C}} = \begin{bmatrix} \mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{E}} = \begin{bmatrix} \mathbf{E} \\ 0 \end{bmatrix} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix}$$

$$u(t) = -\mathbf{K}\mathbf{x}(t) - k_{n+1}x_{n+1}(t) = \bar{\mathbf{K}}\,\bar{\mathbf{x}}(t)$$

$$\bar{\mathbf{K}} = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}$$

$$\bar{\mathbf{A}} - \bar{\mathbf{B}}\bar{\mathbf{K}} = \begin{bmatrix} 0 & 50 & 0 \\ -200 - 200k_1 & -200 - 200k_2 & -200k_3 \\ -1 & 0 & 0 \end{bmatrix}$$

$$\left| s\mathbf{I} - \bar{\mathbf{A}} + \bar{\mathbf{B}}\bar{\mathbf{K}} \right| = s^3 + 200(1+k_2)s^2 + 10{,}000(1+k_1)s - 10{,}000k_3 = 0$$

$$s^3 + 320s^2 + 6{,}200s + 60{,}000 = 0$$

$$k_1 = -0.38 \; k_2 = 0.6 \; k = -6.0$$

$$\begin{bmatrix} \Omega(s) \\ I_a(s) \end{bmatrix} = \frac{1}{\Delta_c(s)} \begin{bmatrix} -\frac{1}{J}\left(s^2 + \frac{R}{L}s + \frac{k_2}{L}s\right) & -\frac{k_3 K_i}{JL} \\ -\frac{1}{J}\left(-\frac{K_b}{L}s - \frac{k_1}{L}s + \frac{k_3}{L}\right) & -\frac{k_3}{L}\left(s + \frac{B}{J}\right) \end{bmatrix} \begin{bmatrix} \frac{T_L}{s} \\ \frac{1}{s} \end{bmatrix}$$

적분제어를 포함한 상태피드백

$$s = -300, -10 + j\,10, \text{ and } -10 - j10$$

$$\begin{bmatrix} \Omega(s) \\ I_a(s) \end{bmatrix} = \frac{1}{\Delta_c(s)} \begin{bmatrix} -\frac{1}{J}\left(s^2 + \frac{R}{L}s + \frac{k_2}{L}s\right) & -\frac{k_3 K_i}{JL} \\ -\frac{1}{J}\left(-\frac{K_b}{L}s - \frac{k_1}{L}s + \frac{k_3}{L}\right) & -\frac{k_3}{L}\left(s + \frac{B}{J}\right) \end{bmatrix} \begin{bmatrix} \frac{T_L}{s} \\ \frac{1}{s} \end{bmatrix}$$

$$\lim_{t\to\infty} \begin{bmatrix} \omega(t) \\ i_a(t) \end{bmatrix} = \lim_{s\to 0} s \begin{bmatrix} \Omega(t) \\ I_a(t) \end{bmatrix} = \begin{bmatrix} 0 & K_i \\ 1 & B \end{bmatrix} \begin{bmatrix} T_L \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ T_L \end{bmatrix}$$

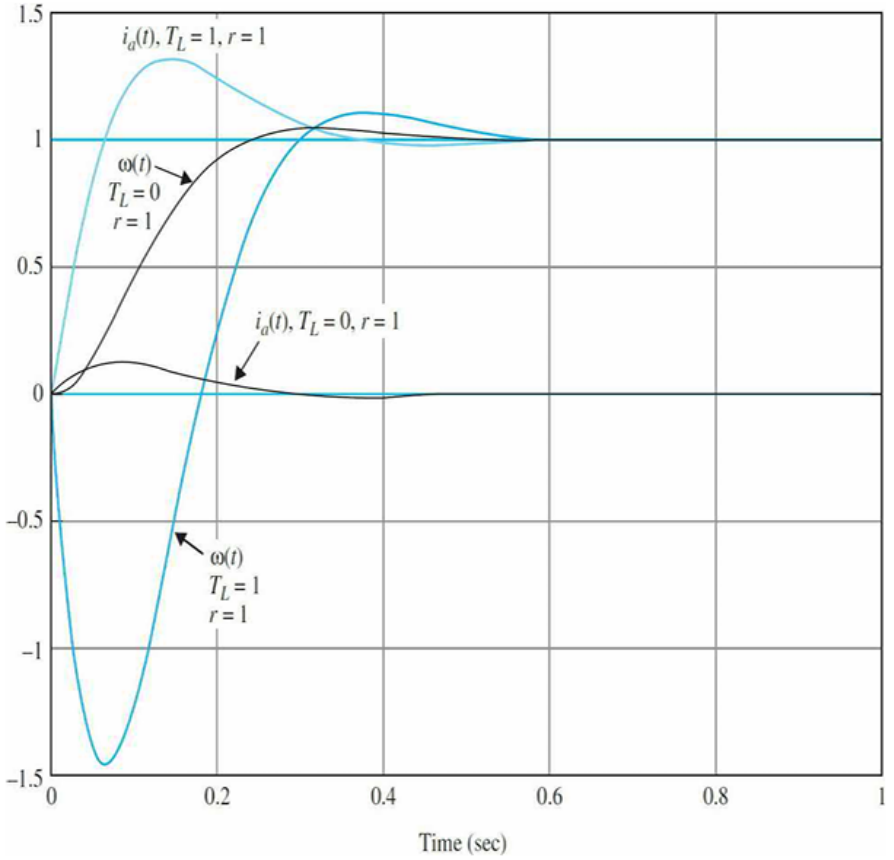


Figure 8-30 Time responses of dc-motor control system with state feedback and integral control and disturbance torque in Example 8-18-2.